

Business Intelligence Solution for Covid-19 Pandemic

Mozah AlDhanhani

*Artificial Intelligence Research Center (AIRC),
College of Engineering and Information Technology
Ajman University, Ajman, United Arab Emirates*

Sharif Naser Makhadmeh

*Artificial Intelligence Research Center (AIRC),
College of Engineering and Information Technology
Ajman University, Ajman, United Arab Emirates
s.makhadmeh@ajman.ac.ae*

Mohammed Azmi Al-Betar

*Artificial Intelligence Research Center (AIRC),
College of Engineering and Information Technology
Ajman University, Ajman, United Arab Emirates
m.albetar@ajman.ac.ae*

Abstract—The COVID-19 pandemic, first detected in December 2019, spread drastically globally within a short period. Such a pandemic has caused enormous loss of lives and health complications and adversely affected the world economies. Effective tracking of the spread of the disease around the globe and predicting when the next wave will occur has become critical in measures geared towards mitigating COVID-19. This paper explores different ways of utilizing analytics and business intelligence tools and solutions to understand the spread of COVID-19 around the globe and predict the number of new COVID-19 cases likely to be recorded. Microsoft Power BI is used to visualize COVID-19 data simply and intuitively in different ways for health decision-makers and concerned parties to easily understand the spread of COVID-19 through various visualizations and dashboards. We also utilize predictive analytics capabilities in Microsoft Power BI to predict the number of COVID-19 cases likely to be recorded in the next few months. The obtained results showed that COVID-19 cases increased over time, particularly in crowded countries. In addition, the results proved that the death rate is reducing with time even though the cases number is increasing.

Index Terms—Business Intelligence, COVID-19 Pandemic, Visualizations

I. INTRODUCTION

Data visualization is a vital skill in the business world as it enables executives to formulate decisions quickly given the needed information. It is a significant part of how business intelligence has evolved to what is termed the era of business analytics [1]. Data visualization tools enable data scientists to convey their results more efficiently, thereby sharing insights with relevant stakeholders. There are different types of visualization tools. Tableau helps in creating maps, charts, and interactive graphs. The three categories of tableau include desktop, public, and learning. Tableau resembles Excel in functionality, but it is easier to construct graphs [1]. QlikView is a data discovery platform for empowering users to formulate quicker, better decisions through speeding up analytics, illuminating

new business ideas, and increasing the precision of outcomes [2].

The Power BI tool helps in visualizing business intelligence data. Also, it can be employed in predictive analytics, reporting, and self-service analytics. The good thing is that using the Power BI desktop can enable the connection to data from different data sources [3]. An end-user platform can prepare reports and convey insights to colleagues. It is a centralized repository from which business users can obtain data for increased sharing efficiency. The tool has numerous advantages, such as self-service capability and interactive content [4]. An analyst can create a report and add drill-down options so that users can see specific data subsets of interest. Moreover, it is possible to control how the data is gathered and presented. Users can also make supported changes. Hence, it is an easy-to-learn tool, and people can train themselves. The free version of Power BI is appealing. Small organizations can benefit from it, but larger ones may require a paid license.

The Power BI tool is covered by the developer's security policies and license. Therefore, if a firm already has other Microsoft products, the developer does not need to perform a thorough security risk analysis. Logging in can be done conveniently using the existing credentials [4]. It is not necessary to wait for screams; the analysts can view usage metrics. In doing so, it is possible to maintain solutions and remove or change others. The tools can be easily integrated with different big data sources to gather and analyze large amounts of data. The Power Map and Power Query tools can be integrated with Big Data analytics through the Office 365 suite. Map startups help plot geographic data and analyze it in a 3D view. It is possible to get extra data in Excel and then use or break it down appropriately with the query.

Regarding decision-making, data visualization helps by improving accuracy, empowering collaboration, simplifying communication, and increasing speed. Data gathered from a given evaluation and organized into a colourful bar graph can

help managers pick the aspects that require improvement [5]. They can allocate resources to improve such under performing aspects in a few minutes. Information is also simplified, thereby lowering the need to fill the gaps with personal biases. Decisions to allocate resources, as shown above, do not have to be based on presumptions but factual data. The visual approach to presenting information can enable managers to make sure the adjuster is aware of the reasons he or she needs training.

II. RESEARCH MOTIVATION AND OBJECTIVES

Technological solutions have been emerging in the market. However, big data issues hinder the efforts of digital change, thereby leading to poor decision-making. Only a few firms are competing in analytics or steering innovation using data. Majorities claim culture, people, and business processes contribute to 92% of the biggest data challenges [6]. The remaining 8% is due to technological limitations. The implementation is marred by a skills shortage because the supply of analytic and data science skills is lower than the demand. With such a skills shortage, it is hard to take advantage of data. The other problem is poor data quality and data silos. Big data is difficult to consolidate, especially when security issues, external and internal hindrances, and inconsistencies.

The lack of a central accountability point makes analytics a poorly focused endeavour. It can cause misleading decisions and the omission of steps when executed by unprepared teams or individual firms [6]. The absence of a coordinator leads to the failure of the governance strategy. Understanding the data becomes complicated if the implementation approach is disjointed. The firm lacks data assets' visibility, gets incorrect outcomes from algorithms and encounters more privacy and security threats. Money goes to waste due to a lack of value addition since no one takes liability.

Some firms undertake big data projects without investing time in figuring out the exact problem they intend to solve. This haste causes delays to many projects. A big data project should avoid launching with a vague goal. Otherwise, the resultant information would just be a non-sticking heap. The information may even become analysis paralysis and be unable to guide proper decision-making. Choosing easier-to-track metrics is a recipe for project failure. This leads to a waste of resources on non-impactful projects.

Digital transformation has caused an increase in the pace of processing big data. The COVID-19 pandemic has also increased the need to process large amounts of data quickly. Such speed may hinder the ability to adapt. Due to behavioural transformations, the pandemic has made business presumptions and historical data useless. Building a model on pre-pandemic data is recommendable if the available data is inadequate for big data analytics. Specifically, for COVID-19 data, a Power BI model can help to analyze the large data sets from the World Health Organization (WHO) [7], [8].

This study aimed to utilize Microsoft Power BI as a data visualization tool. Power BI is appropriate for analyzing large data sets. Therefore, it was chosen to analyze COVID-19

dataset. The power BI desktop can be assigned to a premium capacity. The data was obtained from the WHO database. Therefore, the main objectives of this study are (i) to analyze and visualize COVID-19 data through data cleaning, transformation, and integration, (ii) use machine learning approaches to predict future cases of COVID-19, and (iii) build a dashboard using Microsoft Power BI to enable decision makers to understand COVID19 data and make decisions faster.

III. LITERATURE REVIEW

Many studies have been performed to establish the application of Power BI. This section provides a review of literature. It covers literature on Power BI as a data visualization tool and its application in COVID-19.

A. Data Visualization Using Power BI

Amrapali et.al. [9] conducted a study to construct a BI solution in line with a self-service BI solution, known as Microsoft Power BI. The study's theoretical component provides an understanding of self-service BI and BI concepts, structures, and relationships [9]. It demonstrates Power BI's use (implementation) in building an effective BI solution that meets the functional requirements for business analytics [9]. This study is significant as it can support and guide the application of Microsoft Power BI to different data visualization projects. Bharanidharan et.al. [10] showed how a dataset could be easily changed by Power BI into dashboards and analytical reports that can be shared. The authors found that Power BI can help research professionals and institutions meet their data analysis requirements [10]. The tool was also found as a radical approach to easing data analytics and business intelligence.

In another study, Debrus et.al. [11] applied Power BI to designing modernized, interactive dashboards for improving data access, review, and interaction. The authors were motivated by the rising significance of data visualization in clinical studies and decision-making. The authors developed different tools to monitor, analyze, and test study data in real-time throughout the research, proving that power BI is suitable for meeting the requirements of medical and clinical teams. Because of the importance of visualization and interpretation of data in the business sector. Michael et.al. [1] conducted a study to determine the kinds of skills employers require when recruiting and compared them with the capability to apply the technical skills in the curriculum of business schools. To achieve their objective, the authors surveyed different software tools, including Microsoft Power BI, IBM Watson Analytics, Tableau, and Excel. They created and evaluated how each software package was perceived in terms of the experience of researchers with dashboard creations. The results revealed that business schools should prepare their students for data analysis and analytics to match their skills with the current business needs. The curriculum should concentrate on teaching data visualization for strategic decision-making.

B. Power BI for COVID-19

Bartosz et.al. [12] examined techniques and tools for data visualization and showed how to integrate them with practical business usage. One of the tools they examined was Microsoft Power BI, and they presented an example of applying this tool in analyzing COVID-19 associated data. The data was to be applied later in the prototype of the mobile application to examine the epidemiological conditions and travel restrictions across the world. Steven et.al. [13] mapped COVID-19 data in Los Angeles with Power BI. The tool helped them obtain dynamic, high-quality data visualization trends of COVID-19.

Matthias et.al. [14] studied the patterns of COVID-19 in Nigeria and analyzed their data using Power BI. The tool helped them visualize some important points and present the analyzed results in plot diagrams. Their ultimate goal was to provide analyzed and visualized data that could help their government know how to prevent the spread of the coronavirus. Selwa et.al. [15] investigated and assessed Moroccan distance learning during the corona pandemic. The authors used Microsoft Power BI to analyze and visualize data and draw insights. They found that online learning was less interesting than the usual learning, thereby calling for means to ensure 50% of physical learning.

Overall, the past literature reveals that Power BI is an effective data visualization tool. Its use during the COVID-19 pandemic has helped ease the data burden, promoted the speed of analysis, and provided accurate outcomes that guide management, treatment, and curb the spread of the disease. Therefore, there is enough evidence to prove the efficiency of Power BI in the visualization of COVID-19 data. As a result, its use in this research is justified.

IV. PROPOSED RESEARCH METHODOLOGY

A. Tools Components

Before using this tool, analysts should download the Power BI Desktop app and sign up for the service. The next step would be to extract data from sources, specifically the World Health Organization, having the tool in place. The second step is data cleaning using Python in Google colab. After running all the commands, the data will be cleaned and replaced in the same file. Data should be shaped with queries that construct insightful, persuasive data models. The third step is to import it into the Power BI desktop. The data should then be uploaded to the workspace. The final step is to create a visualization using a pie chart, a geographical chart, or a bar chart, depending on the user's choice. Afterward, the visualization can be created by choosing columns, finding insights, and finally producing visualizations. The destination for publishing should then be selected. Once the workplace is found, one only needs to click on the "select" button to publish. The report files can be shared so that others can build upon, share, and leverage them. The best means of sharing report files is by uploading them to the Power BI service. The public or other interested parties can access the shared report to learn more about matters surrounding COVID-19.

An interactive line chart is used to show the progression rate of daily deaths and cases to map the daily conformed cases and death. The chart also displays trends, Trends talks about the growth of progression rate.

Estimated active cases are presented using a line chart. To calculate the estimated recovered cases using Data Analysis Expressions (DAX). First, create a measure that will store the X days to recover with the expression (Estimated Days to Recover = 17). Then, create the measure for the estimated recovered cases.

The Estimated Active Cases are calculated by subtracting the total confirmed cases from the estimated recovered cases ($EstimatedAC = [Totalcases] - [EstimatedRecovered]$).

To show the total cases of COVID-19 and deaths in different geographic areas of the world, ArcGIS Maps were used. ArcGIS for Power BI is provided by Esri to built-in intelligent mapping and design maps using a variety of base maps, themes, and color settings. First, create ArcGIS account, then to add the map to the dashboard, Select the ArcGIS for Power BI icon from the Visualizations pane.

The bar chart is used to display the total cases by country and region. It is also used to show five-day average cases by country. A five-day moving average is computed as a new measure using (DAX).

B. Machine Learning Components

Machine learning (ML), a subset of artificial intelligence (AI), is a technique for determining a solution to a real-world problem by learning from past data. In the instance of the COVID-19 pandemic, machine learning can forecast new cases and deaths. For this study R visuals were used to forecast cases of COVID-19 for the next ten days. The first step is to download the Time Series Forecasting Chart from Microsoft APP. Then Install R packages manually, using the R console. The required packages are colorspace, fracdiff, ggplot2, graphics, lmtest, magrittr, nnet, parallel, Rcpp, stats, timeDate, tseries, urca and zoo.

C. Business Requirement

Business requirements address the strategic goals of the research. They describe the overall business objective of the study being conducted. This research address the following business requirements:

- To show the impact of Coronavirus (COVID-19) on global health.
- Employ predictive analysis to predict new COVID-19 cases and death
- Dashboard to visually display analytics and predictions of COVID-19.

D. Data source

The processes needed to transform the data to readable information in the BI system shown in Figure 1

In this research, two datasets were used. The first dataset contained details about cases of COVID-19 recorded in 228 countries, while the second dataset contained vaccination data.



Fig. 1. Data Transformation of the proposed BI system

These datasets were collected from the WHO website ("Data At WHO"). The first dataset containing cases of COVID-19 cases contained the following variables, date reported, which indicates the date in which the COVID-19 statistics were reported, the countries in which the statistics were reported, the WHO region the statistics were reported from, the new COVID-19 cases, the cumulative COVID-19 cases reported, the new death reported from COVID-19 and the cumulative death from COVID-19. All these variables are numerical variables except for the country and WHO region. The vaccination dataset contained six variables, including the country in which the statistics were reported, the dataset the statistics were updated, the total vaccinations, the total vaccinations per 100 people, and the total fully vaccinated individuals. The total fully vaccinated individuals per 100 individuals. All these statistics were captured from the 228 countries in the dataset.

E. Data Integration and Cleaning

The data were checked for missing values and a decision was made on how to treat the missing values identified. If missing values are handled, this may lead to inaccuracies and errors in any statistical analysis that is carried out on these datasets.

Missing values can be treated in various ways. The handling of missing values depends on the analysis to be carried out on the cleaned data. If missing values don't have an impact on the final analysis of the data, they are dropped entirely from the dataset by dropping the row; however, if missing values have a huge impact on the analysis to be conducted on the data, the most appropriate way to treat the missing values is to fill them with the mean of the variable containing the missing values. In this research, since we had a few missing values in the WHO coronavirus dataset, the missing values were dropped from the dataset before undertaking the final analysis.

For the vaccination dataset, we replaced the null values of the variables. After dropping the missing values and replacing the null values, the dataset is checked for duplicates, resulting in a less precise analysis of the data if present in any dataset for analysis. No duplicates were identified in the dataset that required to be dropped.

Many values of zero in a dataset create a skewed distribution, resulting in an inaccurate analysis of the data. The dataset is checked for zeros and removed zeros that would have skewed our final analysis.

F. Functional Requirements

A functional requirement describes a function to be provided by the system. The following is a list of the functions that the dashboard will give.

- Daily covid cases, daily death, Persons Fully Vaccinated, Total Vaccination, Mortality Rate, Estimated Active Cases, Estimated Recovered Cases from Jan 2020 to Feb 2022.
- Rich charts and graphs that are easy to understand, including column chart, bar, and line charts.
- An ArcGIS web map which is an interactive display of geographic information, the map shows the COVID-19 cases and deaths for all countries.
- Functions to filter dashboard elements to enable the search based on country, date, and year.
- Prediction of the new cases and new death using Microsoft R Application.
- The ability to export data into PDF.

G. Non-Functional Requirements

Non-Functional Requirements are specification that describes the system's operational capabilities and constraints. They serve as contains during the development of any system; these attributes might include security, performance, reliability, availability, and usability.

- Interactive Visualization: This allows users to interact directly with the BI application to display visualizations such as plot, graphs and metric that serve their business requirements.
- Performance: The refresh rate and response time of the visualization should be fast enough especially when generating new visualizations or drilling down to particular details in a dashboard. Users should experience swift response from the dashboards with minimal computational resources such as memory and CPUs
- User-friendly: The dashboards should provide a simple and intuitive user interface that users can interact with easily.
- Portability: User should be able to access the system using (Windows, Linux, Mac, Android, IOS) operating systems, installed on any hardware.
- Reliability: The system should have in built fault tolerant mechanisms that prevent system failures at critical times.
- Availability: the system should have a high availability of over 99.99% to ensure users are able to access the system whenever they need it.
- Scalability: The system should be able to scale and comfortably accommodate new dashboards, visualizations without impacting the system performance.

V. RESULT AND DISCUSSION

A. Dashboards

The dashboard which is a single page presentation of visualizations is also called a canvas, the dashboard is used to tell a story and present the most important elements from the data in the simplest and most intuitive manner for users to be able to understand with a simple glance. Dashboards can be customized to meet the business objective of its users, embedded in other applications such as web pages and shared with relevant stakeholders. Figure 2 presents the dashboard

containing the most important statistics and visualizations generated from the COVID-19 dataset.

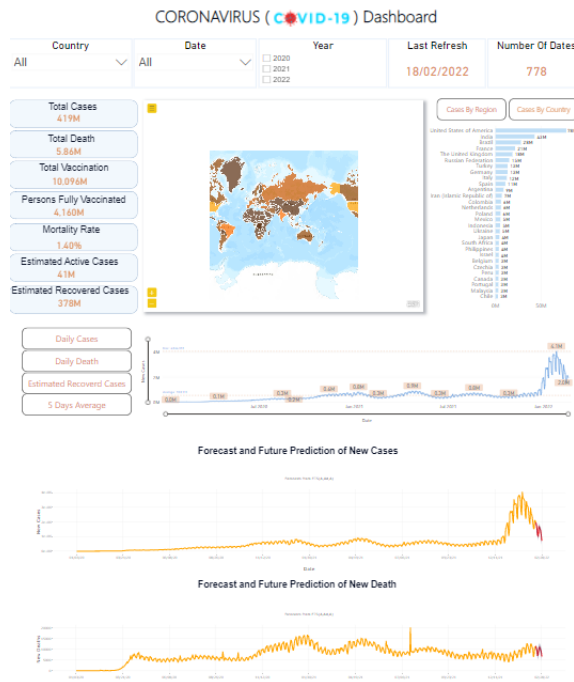


Fig. 2. Dashboard of the proposed BI system

1) *Daily covid cases* : A gradual increment in covid cases is identified with several seasonality peaks within the time frame. The highest covid peak was in the month of January 2022, with over 4 million cases being reported in a single day, as shown in Figure 3.



Fig. 3. Daily covid cases from Jan 2020 to Feb 2022

2) *Daily Death*: The death rate gradually increased from January 2020 to February 2022, with seasonality marching that of the daily new cases. With the increase in daily new cases of COVID-19, the daily death cases also increased. However, this was not the case in the highest peak of new cases recorded in January 2022, as the death rate was not the highest for this period, as shown in Figure 4. This could be because of high vaccinations against COVID-19 .

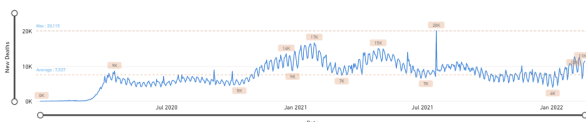


Fig. 4. Daily death from Jan 2020 to Feb 2022

3) *ArcGIS Map visualizations*: Maps and Geographical information systems (GIS) allow organizations to visualize areas experiencing disasters and catastrophes and closely monitor the spread of diseases within a region. This facilitates a faster response to disasters and accelerates the return to normalcy and reopening process in areas that have experienced a catastrophe or disaster.

The ArcGIS map in Figure 5 shows COVID-19 cases and deaths per country. The number of cases in different geographic regions and countries is color-coded as shown on the figure, Light colors have a higher number of cases while darker colors have low cases.

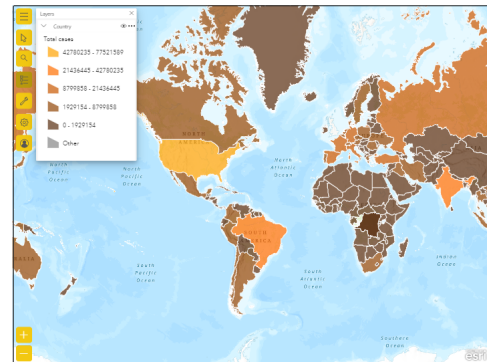


Fig. 5. ArcGIS Map visualization

4) *Total Cases Vs Estimated Recovery*: The trend of estimated recoveries closely mirrors the total daily cases of COVID-19; this trendline indicates that most people who got infected with COVID-19 recovered from the disease, as highlighted in figure 6. When the total cases increase, the estimated recoveries also increase.



Fig. 6. Total cases vs Estimated Recovery from Jan 2020 to Feb 2022

5) *Total Cases By Country*: The data for the total cases shows the top 3 countries with the highest cases, including the United States of America, with 77,521,589 cases reported, India with 42,780,235 cases, and Brazil with 27,806,786 cases. Tuvalu, Turkmenistan, Tokelau, saint Helena, Pitcairn Islands, Niue, Nauru, Micronesia, and the Democratic people's republic of Korea had zero cases reported between January 2020 to February 2022, as shown in Figure 7.

6) *Total Cases Regions*: WHO has categorized countries into regions: Europe, the Americas, South-East Asia, Western Pacific, Eastern Mediterranean, and Africa. The regions were analyzed for cases. Europe has the highest number of cases recorded with 170,033,672 followed by the Americas with 144,596,557, and Africa had the lowest number of reported cases with 8,264,647 cases.

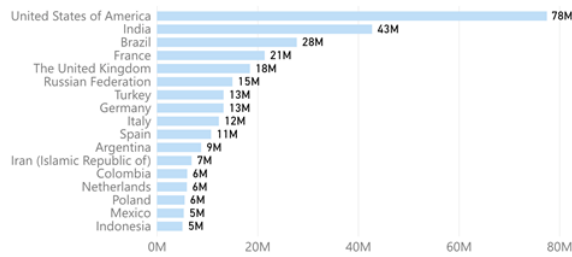


Fig. 7. Total Number of reported cases by Country

7) *Vaccinations*: The total vaccinations for the 228 countries between January 2020 and February 2022 were 10,095,615,243 for the fully vaccinated individuals. As defined by the WHO, 4,159,271,937 individuals were fully vaccinated.

8) *Active Cases* : During the same period, an estimated 378 million individuals recovered from COVID-19, with an estimated 41 million still suffering from COVID-19.

9) *5-days average* : A five-day average is shown in Figure 8. It is also computed as a new measure using DAX. The highest average is 182,265.60.

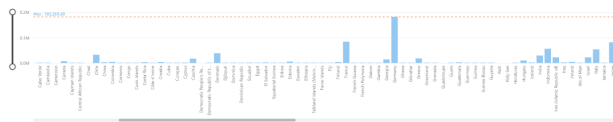


Fig. 8. 5-Day Average Cases

10) *Forecast and Future Prediction*: Figure 9 shows 10-days ahead forecasted result of daily new cases where the daily new cases are predicted to decrease. The total cases are predicted to decrease to 720023 with 95% confidence. The daily new deaths are also predicted to decrease. According to the forecasting, the total deaths will decrease to 9056 with 95% confidence.

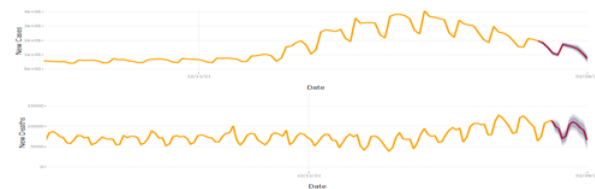


Fig. 9. 10-day Forecasting

VI. CONCLUSION

This study provided data cleaning, analysis, and visualization capabilities through python and Power BI. The analysis and visualizations were based on the number of recorded COVID-19 daily cases, COVID-19 deaths, ArcGIS Map visualizations of COVID-19 cases, Total cases by country, Vaccinations, active COVID-19 cases, and ten-day COVID-19 Forecasting. From the analysis and visualizations in power

BI we can conclude that COVID-19 cases had an upward trajectory from the first cases identified in January 2020 to February 2022, with the peak number of cases being recorded in January 2022, there were also cycles in the number of cases recorded with high peaks in the number of cases at specific times, which start gradually reducing before rising again, this might be attributed to the various strains of covid-19, which cause the spike in the number of cases at various time periods. Though COVID-19 had a low mortality rate of 1.40%. The high number of individuals infected with COVID-19 makes the number of mortalities to be quite high.

REFERENCES

- [1] Michael Diamond and Angela Mattia. Data visualization: An exploratory study into the software tools used by businesses. *Journal of Instructional Pedagogies*, 18, 2017.
- [2] Claire Costa. 8 best data visualization tools that every data scientist should know, 2022.
- [3] Microsoft. Getting started with power bi desktop, 2021.
- [4] Kate Tuzov. Pros and cons of microsoft power bi – an analyst's perspective, 2022.
- [5] Dave Ingram. 4 ways data visualization can improve your decision making, 2021.
- [6] New Vantage Partners. Big data and ai executive survey 2021, 2021.
- [7] Ghada A Altarawneh, Ahmad B Hassanat, Ahmad S Tarawneh, Ahmad Abadleh, Malek Alrashidi, and Mansoor Alghamdi. Stock price forecasting for jordan insurance companies amid the covid-19 pandemic utilizing off-the-shelf technical analysis methods. *Economies*, 10(2):43, 2022.
- [8] Ahmad B Hassanat, Sami Mnasri, Mohammed A Aseeri, Khaled Alhazmi, Omar Cheikhrouhou, Ghada Altarawneh, Malek Alrashidi, Ahmad S Tarawneh, Khalid S Almohammadi, and Hani Almoamari. A simulation model for forecasting covid-19 pandemic spread: Analytical results based on the current saudi covid-19 data. *Sustainability*, 13(9):4888, 2021.
- [9] Amrapali Bansal and A. K. Upadhyay. Microsoft power bi. *International Journal of Soft Computing and Engineering (IJSCE)*, 7(3):2231–2307, 2017.
- [10] Bharanidharan S. and Krishnamoorthy G. Krishnan Vijay. Research data analysis with power bi. *11th International CALIBER*, 2017.
- [11] Roxane Debrus and Terumo Leuven. Power bi in clinical data exploration, powerful visuals that answer relevant questions, 2020.
- [12] Bartosz Sawik and Julia Płonka. Project and prototype of mobile application for monitoring the global covid-19 epidemiological situation. *International Journal of Environmental Research and Public Health*, 19(3):1416, 2022.
- [13] Steven M Graves and Lauren He. Covid-19 mapping with microsoft power bi. *Terra Digitalis*, 2020.
- [14] D Matthias and C Managwu. Data analytics and visualization of covid-19 pandemic in nigeria using power bi.
- [15] Selwa Elfirdoussi, Mohamed Lachgar, Hind Kabaili, Abdelali Rochdi, Driss Goujdami, and Larbi El Firdoussi. Assessing distance learning in higher education during the covid-19 pandemic. *Education Research International*, 2020, 2020.